	$a = 3.2 \times 10^{10}$ m
19	**B** In order to hold the positively-charged particle P halfway between the two plates, the upper plate must be at a lower potential with respect to the lower plate (so that electric force acts upwards which is balanced by the gravitational force acting downwards). By decreasing *V* and causing the positively-charged particle P to move towards the lower plate (which is at a higher potential), the EPE of the particle increases (since a positive charge's EPE increases with electric potential). However, as the particle has fallen to a lower position, its GPE decreases.
20	**C**